Pixel Scan Test Guide — development build 0.1.8 (build 9)

> **Build 9** (0.1.8) is the current build. What changed in each build, and what happened on the Pixel, is listed under *Build history* at the end.

For the project team only. Goal: photograph a label on your Pixel, confirm what was read, and reach the recall-check result **without typing the NDC**. Record what actually happens.

1. Install

Use the attached **`medluma-dev-v0.1.8-b9-arm64-v8a.apk`** (33.3 MB). Universal fallback: `medluma-dev-v0.1.8-b9-universal.apk` (71.2 MB). Full steps (download, checksum, install, Play Protect prompt): `docs/INSTALL_ON_ANDROID.md`.

File	SHA-256
medluma-dev-v0.1.8-b9-arm64-v8a.apk	`be575247ac8b3e90f6893e04b1c3431e86fda6a94e22e3fb3389cb398883241a`
medluma-dev-v0.1.8-b9-universal.apk	`d2481cc0a01064ee1f4236363b45df06fb84030c4f06a40b0f38bc6f5dc1d3fc`

On the Mac:

```
shasum -a 256 ~/Downloads/medluma-dev-v0.1.8-b9-arm64-v8a.apk
```

It installs over builds 1–8. Open **MedLuma (DEV)** → *About and privacy* should show **version 0.1.8 (build 9)**.

If a scan still fails, the error screen now shows a grey **Technical detail (development)** line — please send a screenshot of it (test labels only); it contains no label content.

2. Test labels

Open the attached **`scan-test-labels-SYNTHETIC.pdf`** on your Mac screen (zoom so one label fills most of the screen) or print it. Every page is a synthetic label with real public drug codes and the expected result. Turn down screen glare; hold the phone 15–25 cm away.

You may also try a real label of your own; everything stays on the phone. For a round pharmacy vial: fill the frame with the label (about 10–15 cm away), keep the drug-name line flat toward the camera, and avoid glare across the text. If only some details are read, the form opens prefilled with them and a note "We read some details from your photo…" — fill in the rest. Do **not** send screenshots of real labels to anyone.

3. Steps for each label

1. Home screen → **Scan medication label**. The Pixel camera opens.
2. Take the photo, tap the ✓ (accept) in the camera.
3. "Reading your label on this phone..." (first scan may take a few seconds while the model loads).
4. On *Is this your medication?* check the three groups:
    - **Matched in FDA records** — product (NDC), ingredient/strength, form, labeler.
    - **Read from your label — please check** — e.g., a lot number.
    - **Needs your input** — anything not found.
5. Switch on each item, answer the release question if shown, tap **Yes, these details match my label**.
6. Note the result label and headline.

At the bottom of the confirmation screen, a grey line "Scan details (development): ... ms" shows the on-phone timing — please write it down.

#	Label (PDF page)	Expected (verified with live FDA data on 2026-11-12 using the same OCR code on a computer)
S1	Pharmacy label, metformin ER, NDC printed	Product 62756-142-02 under *Matched in FDA records*; result **Possible recall match** (no lot on pharmacy labels)
S2	Pharmacy label, rosuvastatin	Product 82009-018-10 matched; result **No matching recall found in the FDA records checked** with the date (or a newer FDA result)
S3	Pharmacy label, warfarin	**We don't check this medication** — no recall check
S4	Pharmacy label without NDC, metformin ER	Fields under *Read from your label*; product "Not identified"; **Possible recall match**
S5	Pharmacy label, sertraline	**Not covered by this version**
S6	Manufacturer bottle, metformin ER, lot JKU4639A	Product matched; lot under *Read from your label*; switch on the lot → **Recall match found**, D-0325-2021
S7	Carton with 2D barcode	Product, lot and expiry from the barcode; → **Recall match found**

4. Other checks

#	Check	Expected
C1	Cancel in the camera	Back on Scan with "No photo was taken."
C2	Photo of a blank wall	"We couldn't read your label" with *Scan again* / *Type the details instead*
C3	After S1, tap **Correct the details**	Form prefilled with the NDC, name and strength read
C4	Open Google Photos after scanning	**No** scan photos appear
C5	Airplane mode, then scan S2	Text is read on the phone; then "We couldn't reach FDA's data. Nothing was checked."
C6	Rotate the label ~10°, dim light	Note whether it still reads; record the timing
C9	Real vial of your own (optional)	Note which details were read (no screenshot needed); if "couldn't read", note the *Technical detail* line — it contains only counts

#	Check	Expected
C7	Back button on any result	Returns to the Scan screen
C8	*Type the details instead*	Manual form still works (fallback)

4a. Result screen (build 7)

Use S1 (possible match), S2 (no match) and S6 (recall match found) from the test sheet, or the typed test medications in `docs/FUNCTIONAL_TEST_PLAN.md` (D1, D2, D4).

#	Check	Expected
R1	Top of any result	Label, headline, then one sentence "We checked N FDA recall records for ...: ..." counting each group
R2	Directly below the result box	**Important limits of this result** with the limits and the standing disclaimer
R3	S6 / D1 (recall match found)	**Recalls that match your product** shows D-0325-2021 as a card: recall started, recalling firm, reason, FDA class in plain words, status in plain words, closing date, recall number, *Official FDA record*
R4	S1 / D2 (possible match)	**Recalls that might match**: three cards, then "Show N more records that might match" opens the rest in the same section; "You did not enter a lot number" in the result box
R5	S2 / D4 (no match)	Label **No matching recall found in the FDA records checked**; the words "active recall" appear nowhere
R6	**Records for a different product**	Counts by reason that add up to the number in the summary; no records listed until *Show these N records*; each opened card starts with "Why it does not apply..."
R7	*More about this record* on any card	Product text and lot codes as published by FDA
R8	Status note	Each record section starts with "...A completed or terminated recall does not mean the product is safe..."
R9	Airplane mode, then a check	"We couldn't reach FDA's data. Nothing was checked." — no summary, no counts
R10	Largest font size	All of the above readable; buttons reachable by scrolling

5. Record results

#	Pass/Fail	What the screen said	Scan time (ms)	Notes
S1				
...				

Also note: Pixel model, Android version, which APK (ARM64/universal).

If the app closes by itself, with the phone connected to the Mac:

```
adb logcat -b crash -d > ~/Desktop/medluma-crash.txt
```

That file contains only crash traces (the app logs no label text). If **Scan medication label** is missing and you only see the typed form with "Scanning is not available on this phone yet", the native scanner module was not found at run time — report that; it is the main untested risk.

Build history

> **Build 9** is build 8 with one wording change: the development banner now says "for internal testing only".

> **Build 8** is build 7 with smaller text-reading models (APK 41.4 → 33.3 MB). Screens and results are unchanged. Please check that scanning still reads labels as well as build 7 did, and note the scan time shown in the development detail line.

> **Device result for build 6 (team's Pixel, 2026-11-12):** a label was scanned and the recall check completed (no matching recall), but the result listed every FDA record checked in full. **Build 7** keeps the scanner unchanged and redesigns the result screen to put a summary first (section 4a below).

> **Device result for build 4 (team's Pixel, 2026-11-12):** every scan ended with "We couldn't read your label — The photo could not be read." Cause: the barcode library's `read()` was called with a null crop rectangle, which its Kotlin API rejects, so every analysis threw. Fixed in build 5; text and barcodes now fail independently, and development builds show a "Technical detail" line on this screen. Please use build 5.

> **Device result for build 5 (team's Pixel, 2026-11-12):** no crash, but every photo — the test labels and a real pharmacy vial — ended with "We could not find any text or barcode… Technical detail (development): 0 text lines, 1 barcodes". The barcode reader worked; the text reader returned nothing. Cause (reproduced on a computer): the app took the OCR character list from ONNX Runtime's metadata API, which does not return the 6,623 entries the recognition model needs (on a computer it returns 6,621 and garbles one character; on the phone evidently nothing usable), so every text line decoded as empty and was silently dropped. **Build 6** reads the list directly from the model file, checks its size against the model (a mismatch now shows a TEXT_FAILED technical detail instead of "0 text lines"), and also:
>
> - uses the newer PaddleOCR PP-OCRv4 mobile models;
> - reads text from a 3200-pixel copy of the photo (was 1600), because on the real vial photo the print was only ~15 pixels tall;

- keeps a known drug name even when the strength is unreadable (the form then asks you for the strength — it is never guessed), and corrects "S" read for "5" right after a drug name;
- shows OCR counts in the development detail line: "text regions N (empty E, unclear U); dictionary 6623".

On a computer, the same code on a real vial photo taken by the team now reads "AMLODIPINE" and "5 MG" (the photo is not stored in the repository). This is **not** a phone result; please re-test.